

Problem Statement

Traditional digital commerce has been built around keywords, ads, and product catalogs designed for humans browsing websites. However, consumer behavior is rapidly shifting toward **intent-based interactions** through AI assistants such as ChatGPT, shopping agents, and conversational interfaces.

Instead of searching for *"running shoes size 10"*, consumers increasingly ask questions such as:

"I'm training for a half marathon in Singapore's humid weather and need lightweight shoes under S\$200."

Or:

"Find me a sustainable skincare routine for oily skin that takes less than 5 minutes every morning."

AI agents can only recommend products effectively when brands provide content that is **structured, contextual, persuasive, and optimized for machine understanding**. Most brands today still produce content primarily for websites, search engines, and social media, leaving a significant gap between how products are marketed and how AI agents discover, reason about, and recommend them.

The Challenge

Design and build a solution that helps brands create, optimize, or evaluate product content so that it can be effectively understood, selected, and recommended by AI-powered shopping assistants and conversational agents.

Participants should consider questions such as:

- What information does an AI agent need to confidently recommend a product?
- How should brands describe products beyond traditional titles and specifications?
- How can product attributes, customer personas, use cases, comparisons, and storytelling be represented in ways that AI systems can reason over?
- How can brands measure whether their content is "AI-ready"?
- Can generative AI be used to automatically transform existing product catalogs into agent-optimized content?

Expected Outcomes

Teams may build solutions such as:

- An **AI Content Copilot** that generates agent-friendly product descriptions from raw catalog data.
- A **Content Readiness Score** that evaluates how likely a product is to be recommended by conversational AI systems.
- A **Persona-Aware Content Generator** that creates tailored narratives for different consumer intents.
- A **Simulation Platform** that tests products against thousands of natural language shopping queries to identify content gaps.
- A **Structured Knowledge Layer** that converts marketing assets into machine-readable representations for AI commerce applications.
- Novel approaches that help brands improve their visibility and competitiveness in an ecosystem increasingly mediated by AI agents.

Rubric

- 1. Problem Comprehension — Does the team demonstrate a clear understanding of why AI agents struggle to reason about product content, and have they correctly identified the gap they are solving for?
- 2. Solution Architecture — Is the underlying system well-thought, well designed and technical approach justified given the problem?
- 3. AI Reasoning Quality (Based on live demo) — When given real intent-driven queries, does the system surface the right products?
- 4. Scalability & Generalisability — Does the architecture work across different product categories and datasets, or does it only function within a narrow, pre-defined scope?
- 5. Brand Adoptability — Would a real brand actually be able to implement this? Is there a clear integration pathway with minimal friction?